COLLEGE OF CREATIVE INNOVATION

Programme

Master of Software Engineering
(Level 9) 180 Credits

Course

MSE806: Intelligent Transportation Systems
(15 Credits)

Assessment 2

Group Project

Weighting within the course:

50%

Assessment Tasks to Learning Outcome and GPOs mapping

Tasks	Learning Outcomes	GPOs
Task A	LO3	GPO2
Task B	LO4	GPO5

Objective

The objective of this project is to empower students to explore, design, and implement innovative solutions within the Intelligent Transport System (ITS) domain. While the overarching goal is to contribute to advancements in transportation technology, students are encouraged to propose projects that align with their interests and showcase creativity. The assessment will focus on the relevance, feasibility, and impact of the chosen project within the broader context of ITS.

Assessment Instructions

- This assessment is an open book activity, you can use your own course and review notes as well as offline or online resources, such as textbooks or online journals.
- You can always ask your tutor if you need further explanation or if the instructions are not clear
- By completing and submitting an assessment you are authenticating that you are the original creator and/or author of all the submitted work and that it does not violate plagiarism or copyright law. All written work such as essays, reflections, reports must be in your own words. Please refer to the Academic Misconduct and Authenticity of Assessment Evidence statement in your Student Handbook for more information.
- Please ensure the completion of the assessment by **week 15.**
- Grades and feedback will be returned within 15 days of the submission date.

Learning Outcomes

LO3: Apply intelligent transportation technologies, knowledge, and skills in the development of reliable software information communication systems.

LO4: Analyze and assess the efficiency of intelligent transportation systems to support enhanced system security.

Graduate Profile Outcomes (GPOs) covered

GPO2: Utilize highly specialized knowledge and skills to carry out cutting-edge software engineering projects independently and collaboratively.

GPO5: Critically evaluate current cutting-edge research in software engineering and apply it to industry practice.

Success Criteria:

You need to meet all the requirements of each of the learning outcomes and receive 50% or more to pass this assessment. You are allowed a maximum of three attempts. To meet all the requirements of each of the learning outcomes, you must achieve PASS results for each task item.

Grading:

The final grade will be determined by the score achieved in this assessment based on the following table. Should a second or third attempt be required the maximum contribution toward the overall mark for the tasks that required a second or third assessment attempt is 50%. A late submission is considered a second attempt, so the contribution will be capped at 50%.

Grade	Mark Band Range
A+	Meet all course requirements, mark range (90-100)
A	Meet all course requirements, mark range (85-89)
A-	Meet all course requirements, mark range (80-84)
B+	Meet all course requirements, mark range (75-79)
B	Meet all course requirements, mark range (70-74)
B-	Meet all course requirements, mark range (65-69)
C+	Meet all course requirements, mark range (60-64)
C	Meet all course requirements, mark range (55-59)
C-	Meet all course requirements, mark range (50-54)
D	Did not meet all course requirements, mark range (40-49)
E	Did not meet all course requirements, mark range (0-39)

Submission requirements:

- Submit your assignment in a single ZIP folder that includes all necessary files: project report files in DOC or PDF format, code files for the project implementation, and executables or any additional files required for the project demonstration.
- Ensure the submission of your assessment to **Turnitin** and generate a report adhering to the acceptable threshold of 20% or lower for originality.
- Download the files from Turnitin and subsequently submit them to Blackboard. Include the Turnitin report as part of your final submission file.
- Ensure that your report falls within the word limit of 2500 to 3000 words.
- The submission must have your name and ID number clearly printed.
- Submission should be typed, double-spaced, and adhere to a standard citation format if external sources are referenced.
- Employ academic formatting and adhere to the APA citation style for references.
- Submit a recorded presentation file along with your presentation slides in PDF or PPT format. Your combined presentation of Task A and Task B should not exceed 15 minutes.

Assessment Tasks

Task A: Apply Intelligent Transportation Technologies

LO3: Apply intelligent transportation technologies, knowledge, and skills in the development of reliable software information communication systems.

Activities:

1. **Project Proposal:**
    - **Project Title Proposal:** Choose a project title reflecting your passion and interests in ITS.
    - **Define Objectives:** Clearly outline specific project objectives within the ITS domain, addressing challenges or opportunities.
    - **Technological Innovation:** Explore emerging technologies to advance ITS solutions, focusing on reliability and communication systems.
    - **Ethical Considerations:** Prioritize privacy, security, and responsible technology use in software development.

Deliverables:

- Proposal document detailing project title, objectives, technological innovations, and ethical considerations.

Task 2: Analyze and Assess ITS Efficiency

LO4: Analyze and assess the efficiency of intelligent transportation systems to support enhanced system security.

Activities:

1. **Project Implementation:**
    - **Methodology:** Describe methodologies and approaches used in implementing the ITS project.
    - **Results and Analysis:** Present project outcomes, including software functionalities and performance metrics.
    - **Security Assessment:** Analyze ITS efficiency in enhancing system security, evaluating effectiveness and impacts.
2. **Discussion and Conclusion:**
    - **Discussion:** Reflect on findings and implications for enhancing system security within ITS.
    - **Conclusion and Self-Reflection:** Summarize key project achievements, personal contributions, and lessons learned during project implementation.

Deliverables:

- Project report encompassing introduction, literature review, methodology, results and analysis, discussion, conclusion, references along with implemented project files.

Marking Rubrics

LO3: Apply intelligent transportation technologies, knowledge, and skills in the development of reliable software information communication systems.
LO4: Analyze and assess the efficiency of intelligent transportation systems to support enhanced system security.

Criterion	LOs	Weighting	A (80-100) %	B (65-79) %	C (50-64) %	D (40-49) %	E (0-39) %
Report Writing in APA Style	LO3 LO4	10%	Exceptionally clear, well-organized report, with thorough APA formatting. No errors in citations or formatting.	Clear and organized report with minor APA formatting errors.	Report has some organization issues and several APA formatting errors.	Report is poorly organized with many APA formatting errors.	Report lacks organization and has numerous APA formatting errors.
Understanding and Application of Intelligent Transportation Technologies	LO3	15%	Demonstrates exceptional understanding and application of intelligent transportation technologies. Thoroughly integrates knowledge into the project.	Demonstrates good understanding and application of technologies. Mostly integrates knowledge into the project.	Demonstrates adequate understanding and application with some integration of knowledge.	Demonstrates limited understanding and application with minimal integration of knowledge.	Demonstrates poor understanding and application, with no integration of knowledge.
Software Information Communication Systems Development	LO3	25%	Develops a highly reliable and efficient software communication system with innovative features.	Develops a reliable software communication system with some innovative features.	Develops a somewhat reliable system with basic features.	Develops a system with limited reliability and basic features.	Fails to develop a reliable software communication system.
Analysis and Assessment of Intelligent Transportation Systems	LO4	15%	Provides a comprehensive and insightful analysis of system efficiency and security.	Provides a thorough analysis with some insights into system efficiency and security.	Provides an adequate analysis with basic insights into system efficiency and security.	Provides a limited analysis with minimal insights into system efficiency and security.	Fails to provide a meaningful analysis of system efficiency and security.
Project Implementation	LO3 LO4	25%	Implementation is highly effective, demonstrating exceptional planning and execution.	Implementation is effective, demonstrating good planning and execution.	Implementation is somewhat effective, with adequate planning and execution.	Implementation shows limited effectiveness and planning.	Implementation is ineffective with poor planning.
Final Presentation	LO3 LO4	10%	Presentation is exceptionally clear, engaging, and well-structured. Demonstrates deep understanding of the project.	Presentation is clear and well-structured. Demonstrates good understanding of the project.	Presentation is somewhat clear and structured. Demonstrates adequate understanding of the project.	Presentation is unclear and poorly structured. Demonstrates limited understanding of the project.	Presentation is confusing and disorganized. Demonstrates poor understanding of the project.
Total		100%					

Note: The ranges for each grade level encompass the full 11-point grading system as outlined in the accompanying table. Please refer to the table for detailed percentage ranges associated with each letter grade.